Maintenance Procedure MP-07: Hydraulic Test Bench HTB-3

Work instruction, revision D, owner: Maintenance department, valid from 15 January 2026
Rheintal Antriebstechnik GmbH

1. Purpose

This procedure describes the preventive maintenance of the hydraulic test bench HTB-3 in hall 2, used for end-of-line testing of RA-HP pumps. Following it keeps the measurement uncertainty within the limits required by the ISO 9001 test plan and prevents unplanned downtime.

2. Responsibilities and qualification

Preventive maintenance may only be carried out by maintenance technicians who have completed the HTB-3 training module and the annual hydraulics safety refresher. Electrical work on the 400 V drive cabinet is reserved for qualified electricians. The shift supervisor releases the bench for maintenance in the CMMS and signs the lockout tag.

3. Safety: lockout and depressurising

• Switch off the main drive (160 kW motor) at the main switch Q1 and lock it with a personal padlock.

• Open the accumulator discharge valve V12 and wait until the pressure gauge PI-3 shows 0 bar.

• Wait at least 10 minutes after switching off before opening any hydraulic connection; the fluid may be up to 70 °C.

• Wear safety glasses, nitrile gloves and safety shoes. Hearing protection is required while the bench is running.

• Place absorbent mats under every connection that is opened. Spilled fluid is a slip hazard.

4. Maintenance schedule

Interval	Task	Duration
Weekly	Visual leak check, check fluid level in the 1200 l tank, clean the test area	20 min
Monthly	Check filter clogging indicators, test the emergency stop, calibrate PI-1 to PI-4 zero point	1 h
Quarterly	Oil sample for cleanliness and water content, check accumulator pre-charge pressure	2 h
Semi-annually	Replace pressure filter F1 and return filter F2, check hose assemblies	3 h
Annually	Replace all hoses older than 6 years, calibrate flow meter FT-1 and torque sensor, oil change	8 h

5. Detailed steps

5.1 Oil sample (quarterly)

Take the sample while the bench is running at operating temperature from the sampling valve SV-1 in the return line, after flushing 200 ml. Use the laboratory bottles from the stores (article 77-0931). Acceptance limits: cleanliness class ISO 4406 18/16/13 and water content below 300 ppm. If a limit is exceeded, run the offline filter unit for 24 hours and repeat the sample.

5.2 Accumulator pre-charge (quarterly)

The 50 l bladder accumulator must be pre-charged with nitrogen to 90 bar at 20 °C. Check with the charging and testing device RA-FPU after depressurising the hydraulic side. A pre-charge below 80 bar means the bladder is leaking; replace the bladder and record the serial number in the CMMS. Never use oxygen or compressed air for pre-charging.

5.3 Filter replacement (semi-annually)

Pressure filter F1 uses element RA-FE 10/25 (10 micron absolute). Return filter F2 uses element RA-FE 10/60. Lubricate the O-rings with hydraulic fluid before fitting. Tighten the filter bowl by hand plus a quarter turn. After the restart, check the clogging indicator; it must show green at operating temperature.

5.4 Flow meter calibration (annually)

The turbine flow meter FT-1 is calibrated against the reference gear meter at five points between 20 and 200 l/min. The permissible deviation is 0.5 percent of the measured value. Record the calibration in the test equipment database and apply a new calibration label. If the deviation exceeds 0.5 percent, send the meter to the manufacturer; the spare meter from cabinet S4 is used in the meantime.

6. Restart and release

Remove all tools and mats, close valve V12, remove the lockout and switch on Q1. Run the bench for 5 minutes at minimum pressure and check for leaks. Then run the reference pump HP450-REF and compare flow and pressure with the reference values; the deviation must be below 2 percent. Record the maintenance in the CMMS work order and inform the shift supervisor, who releases the bench for production.

7. Records

Record	Where	Retention
Maintenance work order	CMMS	10 years
Oil analysis report	Quality drive, folder HTB-3	10 years
Calibration certificate FT-1	Test equipment database	Until next calibration + 5 years
Hose replacement list	CMMS asset HTB-3	Life of the bench